

Monitoring jobs and run history

Run outcomes, the matrix view, spotting trends against a baseline, and reading a DAG for the upstream blocker.

Monitoring is **comparison**. A single run tells you almost nothing. A run against forty previous runs tells you everything.

The six job run outcomes

OUTCOME	MEANING
Succeeded	All tasks were successful
Succeeded with failures	Some tasks failed, but all leaf tasks were successful
Failed	One or more leaf tasks failed
Skipped	The run never executed — for example, maximum concurrent runs was exceeded
Timed Out	The run took too long and timed out
Canceled	Somebody, or the platform, cancelled the run

ON THE EXAM

Remember from module 17: job outcome is determined by the leaf tasks, the ones with no downstream dependencies.

Succeeded with failures

- ◆ **It is a distinct outcome**, not a variant of Succeeded and not a variant of Failed.
- ◆ **It means something inside the job failed**, and the tasks at the end of the graph handled it.
- ◆ **A job-level failure notification will not fire**, because the run did not fail.
- ◆ **This is exactly the state module 17's cleanup pattern produces**, and it is why per-task notifications matter.

ON THE EXAM

If you filter run history by Failed, you will not see these. Filter for it deliberately, or handled failures stay invisible.

The matrix view

- ◆ **One column per run, one row per task**, so recent history is visible at a glance.
- ◆ **The top row is the run's total duration** and overall state.
- ◆ **Bar height indicates duration.** That is how a trend becomes visible without any calculation.
- ◆ **Hover a cell** for that task's start time, duration, cluster and status.

ON THE EXAM

The matrix answers two questions at once: which task failed, and whether it has failed before. A list view answers only the first.

What the colours mean

COLOUR	STATE
Green	Successful
Red	Unsuccessful
Pink	Skipped
Yellow	Waiting for retry
Grey	Pending, cancelled, or timed out

ON THE EXAM

Pink and red mean different things. A skipped task did not fail; something it depended on did.

The runs list

- ◆ Shows start time, name, type, the identity it ran as, and what launched it.
- ◆ Launched tells you the trigger: a schedule, an API request, or a manual start.
- ◆ Elapsed time for a running job, total time for a completed one.
- ◆ A warning is displayed if the duration exceeds a configured expected completion time.

ON THE EXAM

The 'Launched' column answers a question you will ask during an incident: did somebody run this by hand?

Retention

- ◆ **Databricks retains run history for 60 days**, for both jobs and pipelines.
- ◆ **So a baseline older than 60 days does not exist in the UI.**
- ◆ **For longer history, query the system tables**, which is how you compare against last quarter.
- ◆ **By default the runs table shows failed, skipped and successful runs**, and the filters at the top apply to it.

ON THE EXAM

Sixty days is enough to see a trend and not enough to compare a year-end run against last year's. Plan accordingly.

Reading the DAG for an upstream blocker

- ◆ The job run details page shows the **task graph**, with each task's state on it.
- ◆ Find the **first failure in dependency order**. Everything downstream of it is a consequence.
- ◆ A task can be unsuccessful two ways: it failed, or it was skipped because a dependency failed.
- ◆ **Disabled tasks are marked on the DAG**, either explicitly disabled or disabled because an upstream task was.

ON THE EXAM

The loudest failure is rarely the cause. Follow the arrows backwards until you reach a task that failed with nothing failed before it.

Task-level detail

- ◆ **Click a task to open its run details**, then choose a previous run from the history dropdown.
- ◆ **That gives you a per-task baseline**, which is finer than the job-level one.
- ◆ **A For each task shows a table of its iterations** rather than a single result.
- ◆ **Filter to only failed iterations** when a loop of two hundred has three problems.

ON THE EXAM

A job that is 10 minutes slower is a job-level observation. Knowing which task absorbed those 10 minutes is a task-level one.

Termination reasons worth recognising

CODE	WHAT IT POINTS AT
DRIVER_ERROR	A problem communicating with the Spark driver
CLUSTER_ERROR	The cluster itself failed. Module 26 covers these.
CLOUD_FAILURE	A cloud provider issue, not your code
MAXIMUM_CONCURRENT_RUNS_REACHED	A run was skipped because one was already in flight
REPOSITORY_CHECKOUT_FAILED	The Git checkout failed — a CI/CD problem, not a data one
CLUSTER_TERMINATED_BY_USER	Someone terminated the cluster outside the jobs service

ON THE EXAM

These separate your problems from the platform's, and data problems from infrastructure ones. That is most of triage.

System tables

- ◆ The **system.lakeflow** schema holds **job and task run records** across the whole account.
- ◆ **Query it like any table**, so failure rates and duration trends become SQL rather than screenshots.
- ◆ **Join it to the billing tables** to attribute cost to individual jobs, which needs the tags from module 19.
- ◆ **This is how you exceed the 60-day UI window**, and how you monitor hundreds of jobs at once.

ON THE EXAM

The UI is for one job. System tables are for an estate. The objective's 'failure rates' is a system tables question at scale.

Pipeline monitoring

- ◆ **Every pipeline generates an event log** covering updates, lineage, data quality and resource usage.
- ◆ **The data quality metrics from module 14 live there**, alongside the Data quality tab.
- ◆ **Pipeline runs appear in the same Runs tab**, with the same 60-day retention.
- ◆ **For database connectors, gateway progress** can be monitored through event logs too.

ON THE EXAM

The event log is queryable, so pipeline health can be alerted on rather than watched.

Tiebreakers from this module

WHEN A QUESTION SAYS

The job succeeded but a task failed

Runtime has been creeping up for weeks

A run never executed at all

Many tasks red and pink at once

Compare against last quarter

Failure rates across hundreds of jobs

A slow run that never failed

REACH FOR

Succeeded with failures. Job status comes from leaf tasks.

The matrix view. Bar height shows duration.

Skipped — often maximum concurrent runs exceeded

Find the first failure. Pink means skipped, not failed.

System tables. The UI keeps 60 days.

system.lakeflow, queried

A configured expected completion time triggers a warning

ON THE EXAM

Seven lines. The first and fourth are the two most likely to be tested as 'what does this screen mean'.

A platform team reviews the previous week's job runs by filtering the Runs tab to Failed. Nothing appears, and they report a clean week.

Separately, an analyst reports that a reconciliation table has been missing three regions' data since Tuesday. The job that loads it shows a status of Succeeded with failures on every run since then.

What is happening?

- ☐ A The job status is incorrect and should be reported to support
- ☐ B Succeeded with failures means some tasks failed but all leaf tasks succeeded, and filtering by Failed does not show it
- ☐ C The analyst is looking at a different job from the one the team reviewed
- ☐ D The runs are older than 60 days and have been removed from history

◆ PAUSE AND DECIDE

What is happening?

- A

 The status is incorrect
It is exactly correct. Job outcome comes from leaf tasks, and the leaf tasks succeeded, so the run is not Failed.
- B

 A distinct outcome that a Failed filter excludes
Correct. Succeeded with failures is its own outcome. Filtering by Failed hides every handled failure in the estate.
- C

 A different job
Invents a mix-up. The scenario states the loading job's status directly.
- D

 Older than 60 days
The runs are from this week, and their status is visible, so they are plainly still in history.

THE SIGNAL

The filter created the blind spot, not the platform. A weekly review that only looks for Failed will miss this class of problem indefinitely.

A nightly job has started missing its 06:00 deadline about twice a week. Nobody can say when it started. It has never failed, so no notification has ever fired.

An engineer opens the job's Runs tab and looks at the matrix view for the last month.

What will the matrix view tell her most directly?

- ☐ A Which task is failing, indicated in red
- ☐ B Whether the runtime has been increasing, because bar height indicates duration
- ☐ C The cost of each run, so she can find the expensive one
- ☐ D Nothing useful; she needs to query the system tables instead

◆ PAUSE AND DECIDE

What will the matrix view tell her most directly?

A Which task is failing

Nothing is failing. The scenario says so, which is why no notification fired. There will be no red to find.

C The cost of each run

The matrix shows duration and status, not cost. Cost attribution comes from the billing system tables with tags.

B Whether runtime is increasing, via bar height

Correct. Bar height indicates duration, so a rising staircase across the columns is a visible trend, per task and for the run overall.

D Nothing useful; use system tables

System tables are right for a longer window or many jobs. For one job over one month, the matrix answers it immediately.

THE SIGNAL

'It has never failed' rules out looking for red. The information she needs is encoded in the height of the bars, not their colour.

A job is scheduled every 15 minutes. Reviewing the Runs tab, an engineer sees that roughly a third of the runs show a status of Skipped, coloured pink. The runs that do execute succeed normally.

Recent successful runs have been taking between 18 and 24 minutes.

What is happening?

- ☐ A The job is failing intermittently and the failures are being reported as Skipped
- ☐ B Runs are being skipped because a previous run is still in flight and maximum concurrent runs is 1
- ☐ C The cluster fails to start on those runs
- ☐ D The schedule's time zone is misconfigured

◆ PAUSE AND DECIDE

What is happening?

A Failures reported as Skipped

Skipped means the run never executed. A failure would be red and would carry an error; these have neither.

C The cluster fails to start

That would produce a cluster error and a failed run, not a skipped one. The run would have started.

B A previous run is still running; max concurrent runs is 1

Correct. Runs take 18–24 minutes on a 15-minute schedule, so they overlap, and the default maximum of 1 causes the later trigger to be skipped.

D Time zone misconfiguration

A time zone offset shifts when runs happen. It does not cause a third of them not to happen at all.

THE SIGNAL

The arithmetic gives it away: a job taking 18–24 minutes cannot run every 15. It has outgrown its schedule.

A job with fourteen tasks fails. The matrix view shows one red cell early in the graph and nine pink cells after it. An on-call engineer opens the last pink task, which reports that its dependency did not succeed.

He escalates, saying nine tasks have failed and the job is badly broken.

What should he have done?

- A Repair the run, which will re-run all ten unsuccessful tasks
- B Traced back to the single red task — the first failure in dependency order — since the pink ones were skipped, not failed
- C Opened each of the nine pink tasks in turn to find the common cause
- D Checked the system tables for the failure rate across all fourteen tasks

◆ PAUSE AND DECIDE

What should he have done?

- A

 Repair the run
The right eventual action, and not the right first one. Repairing before knowing why the red task failed just reproduces the failure.
- B

 Trace back to the single red task
Correct. One task failed. Nine were skipped because a dependency failed, which is a consequence, not nine separate problems.
- C

 Open each pink task
Nine tasks will each report the same thing: their dependency did not succeed. Nine identical answers, none of them the cause.
- D

 Check failure rates in system tables
Useful for estate-level trends. It does not tell you why this run failed tonight.

THE SIGNAL

Pink is not red. Nine skipped tasks are one failure with consequences, and the escalation described a crisis that does not exist.

One run is a fact. Forty runs are information. **Find the first failure, and read the height of the bars, not just their colour.**